

RAG Evaluation: RAGAS Metrics, Faithfulness, and Answer Quality

Evaluating RAG systems is more complex than evaluating standard classification or generation tasks because quality depends on both retrieval and generation components. A response can be fluent and well-written while being factually unsupported by the retrieved context. RAGAS (Retrieval Augmented Generation Assessment) is an open-source framework that provides reference-free and reference-based metrics to evaluate all components of a RAG pipeline without requiring human annotations for every query.

1. RAGAS Framework Overview

RAGAS evaluates a RAG pipeline using four primary metrics: Faithfulness, Answer Relevancy, Context Precision, and Context Recall. Each metric targets a specific failure mode. Faithfulness detects hallucination. Answer Relevancy detects irrelevant or incomplete responses. Context Precision detects retrieval noise. Context Recall detects retrieval gaps. Together these four metrics provide a comprehensive view of RAG pipeline health.

RAGAS uses an LLM judge to evaluate each metric. The judge LLM does not need to be the same model as the generator. Using a separate, capable judge model (such as GPT-4 or a large open-source model) reduces bias from self-evaluation. RAGAS 0.4.x returns results as a pandas DataFrame with one row per evaluated sample.

2. Faithfulness

Faithfulness measures whether every claim in the generated answer is supported by the retrieved context. It is the primary metric for detecting hallucination in RAG systems. A faithfulness score of 1.0 means every statement in the answer can be directly traced to a retrieved chunk. A score of 0.5 means half the statements lack contextual support.

2.1 Calculation Method

RAGAS calculates faithfulness by first decomposing the answer into atomic claims using an LLM. Each claim is then independently verified against the retrieved context chunks. Faithfulness equals the number of supported claims divided by the total number of claims. This decomposition approach is more reliable than embedding-level similarity, which may miss subtle contradictions.

2.2 Common Failure Modes

Faithfulness failures typically occur when the LLM generates plausible but unsupported elaborations, mixes information from training data with retrieved context, or makes incorrect inferences from correct

retrieved facts. High faithfulness scores (above 0.9) indicate the generator is staying grounded in the provided context. Scores below 0.7 suggest significant hallucination risk.

3. Answer Relevancy

Answer Relevancy measures how well the generated answer addresses the original question. Unlike faithfulness, which compares the answer to the context, answer relevancy compares the answer to the question. A response that is fully grounded in context but answers a different question than was asked will have high faithfulness but low answer relevancy.

3.1 Calculation Method

RAGAS calculates answer relevancy by generating multiple synthetic questions from the answer using an LLM, then computing the mean cosine similarity between the synthetic questions and the original question. High similarity indicates the answer is focused on the right topic. This reverse-generation approach avoids requiring a reference answer.

3.2 Summary Query Degradation

Answer relevancy scores are systematically lower for broad summary queries. When a user asks 'summarize the main points of this document', a good answer is necessarily long and multi-faceted. The synthetic questions generated from this answer cover many sub-topics, reducing average similarity to the original short query. This is a known limitation of the metric, not a pipeline failure. Separating evaluation by query type (factual vs. summary) is recommended to avoid misleading aggregate scores.

4. Context Precision

Context Precision measures what fraction of the retrieved chunks are actually relevant to answering the question. It evaluates retrieval precision: are the retrieved chunks signal or noise? A score of 1.0 means every retrieved chunk contributed useful information. A score of 0.5 means half the retrieved chunks were irrelevant.

4.1 Relationship to K

Context precision is inversely related to K. Retrieving more chunks increases the chance of including irrelevant content, reducing precision. The tradeoff between K and context precision is a key design decision in RAG systems. Reranking improves context precision by promoting the most relevant chunks to the top of the retrieved set, allowing the pipeline to use a smaller effective context window without sacrificing recall.

5. Context Recall

Context Recall measures what fraction of the ground truth answer's claims are supported by the retrieved context. Unlike Recall@K which operates at the sentence level, Context Recall operates at the claim level using an LLM to decompose the ground truth answer. Context Recall requires a ground truth answer, making it a reference-based metric. A score of 1.0 means all information needed to answer the question correctly was present in the retrieved chunks.

6. Evaluation Dataset Construction

A RAGAS evaluation dataset contains four fields per sample: question, answer, contexts, and ground_truth. The question is the user query. The answer is the pipeline-generated response. The contexts field is a list of the retrieved chunk texts. The ground_truth is a human-written or reference correct answer used for Context Recall computation.

6.1 Dataset Size Recommendations

A minimum of 20 samples is recommended for stable metric estimates. Fewer than 10 samples produce high-variance scores that are not reliable for system comparison. Samples should cover multiple document types and query patterns. For factual queries, ground truth can often be written directly from the source document. For summary queries, ground truth should capture the key points rather than an exhaustive summary.

6.2 Query Type Distribution

A balanced evaluation dataset should include factual queries (specific fact extraction), comparison queries (comparing two concepts), multi-hop queries (requiring information from multiple sections), and summary queries (synthesizing broad themes). A dataset of 20 samples might include 8 factual, 4 comparison, 4 multi-hop, and 4 summary queries. This distribution exercises all retrieval and generation capabilities of the pipeline.

7. Interpreting RAGAS Results

Strong RAG pipelines typically achieve Faithfulness above 0.9, Answer Relevancy above 0.8 on factual queries, Context Precision above 0.75, and Context Recall above 0.8. Answer Relevancy on summary queries is expected to be lower (0.4 to 0.6) due to the metric's known limitation with broad queries. When Faithfulness is high but Context Recall is low, the pipeline is generating accurate but incomplete answers because relevant information is not being retrieved. When Context Recall is high but Faithfulness is low, retrieval is working but the generator is hallucinating beyond the retrieved context.